

Jiseop Byeon

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Education

The University of Texas at Austin, Austin, Texas

Ph.D. in Civil Engineering

Aug. 2026 – Present

M.S. in Civil Engineering

Aug. 2024 – May 2026

- **Advisor:** Dr. Christian Claudel
- **GPA:** 3.96/4.00

Hanyang University, Seoul, Korea

Mar. 2018 – Feb. 2024

B.S. in Architectural Engineering, Minor in Mechanical Engineering

- **GPA:** 4.11/4.50 | **Degree Honors:** Summa Cum Laude
- **Exchange Student** in Civil Engineering, University of Michigan, Ann Arbor (Fall 2021) | **GPA:** 3.73/4.00
- Leave of absence for mandatory military service, Republic of Korea Air Force (Apr. 2019 – Feb. 2021)

Honors and Awards

Birmingham Britton Graduate Fellowship in Engineering, UT Austin

Fall 2026 – Spring 2027

Fulbright Graduate Fellowship, U.S.-Korea Presidential STEM Initiative, Fulbright Korea

Fall 2024 – Spring 2026

Graduate Dean's Prestigious Supplement Fellowship, UT Austin

Fall 2025

George J. Heuer, Jr. Ph.D. Endowed Graduate Fellowship, UT Austin

Fall 2024, Spring 2025

Corporate-sponsored Scholarship, Duckshin EPC Co., Ltd

Spring 2023

Merit-based Scholarship, Hanyang University

Spring 2019, Fall 2021, Fall 2022

Publications

Conference Papers

- [C3] Tianyi Wang[†], Jiseop Byeon[†], Ahmad Yehia[†], Yiming Xu, Jihyung Park, Tianyi Zeng, Sikai Chen, Ziran Wang, Junfeng Jiao, Christian Claudel*, “XR-DT: Extended Reality-Enhanced Digital Twin for Safe Motion Planning Via Human-Aware Model Predictive Path Integral Control,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- [C2] Ahmad Yehia, Abdulllah Mohamed, Tianyi Wang, Jiseop Byeon, Kun Qian, Junfeng Jiao, Christian Claudel, “EgoTraj: Real-World Egocentric Human Trajectory Dataset,” *European Conference on Computer Vision (ECCV)*, pp. 643–660, 2026.
- [C1] Ahmad Yehia[†], Jiseop Byeon[†], Tianyi Wang[†], Huihai Wang, Yiming Xu, Junfeng Jiao, Christian Claudel*, “ARCAS: An Augmented Reality Collision Avoidance System with SLAM-Based Tracking for Enhancing VRU Safety,” *IEEE Intelligent Vehicles Symposium (IV)*, pp. 2038–2045, 2026.

Preprints

- [P2] Ahmad Yehia, Abdulllah Mohamed, Kun Qian, Tianyi Wang, Jiseop Byeon, Omar Hassanin, Christian Claudel, “PAVE: A Cognitive Architecture for Legitimate Violation in Generative Agent Societies,” *arXiv preprint arXiv:2605.19351*, 2026.
- [P1] Christian Claudel, Tianyi Wang, Yiming Xu, Jiseop Byeon, Junfeng Jiao, Javad Mohammadi, Kara Kockelman, Alexandre Bayen, “Solar Phased Arrays-Based Wireless Power for Commercial Aviation Decarbonization,” *Research Square preprint*, 2026.

[†] Equal contribution; * Corresponding author

Presentations

- “XR-DT: Extended Reality-Enhanced Digital Twin for Safe Motion Planning Via Human-Aware Model Predictive Path Integral Control,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, PA, Sep. 2026. (Lightning Talk and Poster)

Research Experience

Mobile Automation and Sensing Systems (MASS) Lab, The University of Texas at Austin Jan. 2025 – Present
Graduate Research Assistant Aug. 2026 – Present

Advisor: Dr. Christian Claudel | In collaboration with Honda Development & Manufacturing of America

- Co-developed XR-DT, an extended-reality digital twin for human–robot interaction with a human-aware MPPI motion planner using egocentric trajectory prediction (IROS 2026).
- Co-developed ARCAS, an AR collision-avoidance system that streams LiDAR-based hazard detections to AR headsets for vulnerable road user safety (IEEE IV 2026).

Skills

- **Languages:** Python, C++, C#, MATLAB, R
- **Libraries / Frameworks:** Numpy, Pandas, PyTorch, Matplotlib, Seaborn, OpenCV, ROS
- **Design / Tools:** Unity, AutoCAD, Sketchup, Rhinoceros 3D / Git, $\text{\LaTeX}$
- **Additional Skills:** LiDAR processing, point-cloud tracking, real-time TCP/IP networking